

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5. HT	(a)			C = 12 H = 1 N = 14 O = 16 (1) 2C = 24 (1) 3H = 3 (1) Mr = 57 24, 3, 14 and 16 seen – 3 marks		4		4	3	
	(b)			Conversion to 30 000 000 g (1) Number = 30 000 000 / 57 (ecf) (1) = 526 000 (1) Answer of 526 = 2 marks 526 320 = 2 marks 526.32 = 1 mark		3		3	3	
	(c)			Indicative content Reaction pathway: - The reaction pathway shown is for an exothermic reaction. - This means that the difference in PE is released as heat. Bond energies: - During a reaction bonds are broken and formed - More energy released when bonds are formed than when bonds are broken Thermal runaway - Thermal runaway occurred because an increase in temperature arose - which causes an increase in reaction rate - which further increases the temperature - The gas pressure was so great that the valve blew open releasing the toxic gas.		6		6		